INFRARED OBJECT DETECTION SYSTEM USING YOLOV8

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PROJECT OVERVIEW

- Problem Statement
- The goal of this project is to build an **object detection system** capable of:
- Detecting objects in infrared images
- Drawing bounding boxes around detected objects
- Classifying each detected object
- Displaying confidence scores for predictions
- Target Classes
- Bicycle
- Car
- People

OBJECTIVES

1. Project Objectives
2. Build a real-time capable object detector
3. Train a custom model on infrared image datasets
4. Improve detection accuracy using data augmentation
5. Evaluate performance using detection metrics
6. Test the model on unseen images

DATASET DESCRIPTION

- **Dataset Information**

- ***Classes Used***
- bicycle
- car
- people
- ***Dataset Features***
- Different lighting conditions
- Multiple object sizes
- Various orientations
- Single and multiple objects per image

Dataset Split	Number of Images
Training Images	295
Validation Images	73
Test Images	108

DATA PREPARATION

- **Data Preprocessing & Annotation**
- Images annotated using bounding boxes
- YOLO format labels used
- Dataset split into:
  - Train
  - Validation
  - Test
- Applied augmentation techniques during training
- ***Model Selection***
- Why YOLOv8?
- We selected **YOLOv8 Nano (yolov8n)** because it offers:
- Fast inference speed
- High detection accuracy
- Lightweight architecture
- Suitable for real-time applications

TRAINING CONFIGURATION

- **TRAINING PARAMETERS**
- **TRAINING SETTINGS**

Parameter	Value
Epochs	50
Image Size	416
Batch Size	2
Augmentation	Enabled
Early Stopping	patience = 10

SYSTEM WORKFLOW

1) Detection Pipeline
2) Load infrared image
3) Preprocess image
4) Run YOLOv8 inference
5) Detect objects
6) Draw bounding boxes
7) Display confidence scores
8) Generate final prediction image

SAMPLE DATASET VISUALIZATION

- Dataset Samples

- Examples from the infrared dataset showing different object classes, lighting conditions, and object scales.

MODEL PERFORMANCE

- Evaluation Results
- Performance Analysis
- High precision indicates low false positives
- Strong recall indicates successful object detection
- Excellent mAP50 demonstrates strong localization capability

Metric	Score
mAP50	0.976
mAP50-95	0.813
Precision	0.962
Recall	0.923

PREDICTION RESULTS

- Model Predictions on Test Images

- Example detections generated by the trained YOLOv8 model with bounding boxes and confidence scores.

EXAMPLE PREDICTIONS

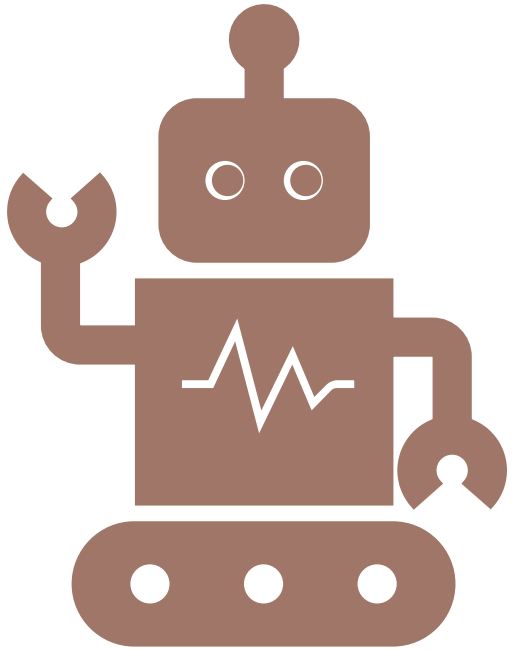
- Sample Outputs
- **Image Example 1**
- car → 0.99
- car → 0.96
- people → 0.90
- bicycle → 0.87
- **Image Example 2**
- car → 0.99
- people → 0.92
- bicycle → 0.91
- Observation
- The model successfully detected multiple objects simultaneously with high confidence scores.

CHALLENGES FACED

- Challenges
- Infrared images have limited texture details
- Small dataset size
- Object overlap in crowded scenes
- Different object scales and orientations
- Solutions
- Used data augmentation
- Fine-tuned YOLOv8
- Applied validation monitoring
- Used pretrained weights

FUTURE IMPROVEMENTS

- Future Work
- Increase dataset size
- Add more object classes
- Improve night-time detection
- Deploy on edge devices
- Real-time video detection integration



CONCLUSION

- Successfully built an infrared object detection system
- Achieved high detection accuracy using YOLOv8
- Model performed effectively on unseen test images
- The system can be extended for real-world surveillance and smart monitoring applications

SCAN
QR_CODE TO
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CODE :